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'graph2graph' neural network?

Graph Summarization (aka Dimension Reduction, Compression)

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Here is a problem that I continue to research on, in my spare time, which I initially started looking at, as a subtopic of my PhD work in the field of Geometric Modeling.

Problem

Similar to Seq2Seq models, are there graph2graph neural networks possible/available?

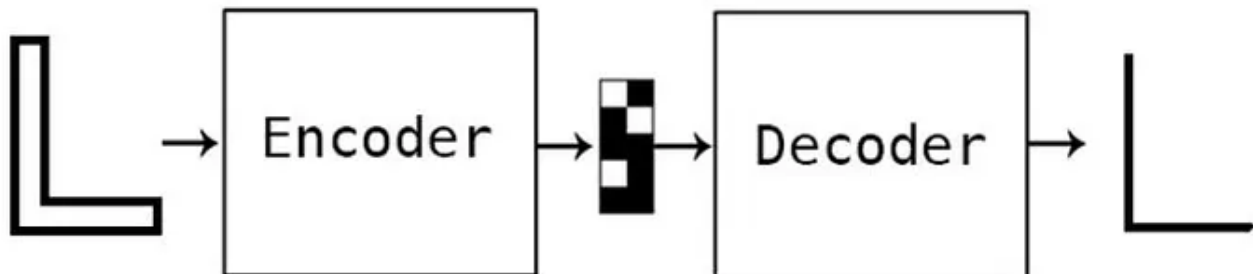
Context

Subtopic of my PhD work was about generating midcurves, a dimension reduction problem on 2D shapes. I attempted it using rules-based programming at that time. Later, as I became familiar with Machine/Deep Learning, thought of attempting it using graph neural networks techniques, where:

- Shapes are represented as a graph.
- Vertices as nodes.
- Connecting curves as edges.
- Dimension reduction operation is called as Midcurve generation.
- Input is 2D profile, say a closed polygon. Example: thick 'L' profile on left in the image below.

- Output is 1D curve in the middle of the profile. Example: thin 'L' curve on the right in the image below.

Wish to build an encoder-decoder network which accepts graphs as input as well as output. I have a training set of such input and output graphs, a supervised set.



As I could not find ready graph2graph network, I converted the problem to image2image (say, pix2pix like) and then solving it as an intermediate step.

More info about current solutions, at my GitHub repo: [yogeshhk/MidcurveNN](https://github.com/yogeshhk/MidcurveNN).

Dataset and code so far, have been open-sourced there. Feel free to work on it, if interested.

My next step is to investigate if graph2graph network is possible or not, how to build such encoder decode network? Is there any such work going on under 'Geometric Deep Learning' or 'Graph Neural Networks'?

Please note that as both, input and output are different, this can not be Autoencoder.

Any ideas?

References

- Vixra paper MidcurveNN: Encoder-Decoder Neural Network for Computing Midcurve of a Thin Polygon, [viXra:1904.0429](https://vixra.org/1904.0429)
- ODSC [proposal](#), [Talk-Video](#)
- CAD Conference 2021, Barcelona, [pages 223–225](#)
- CAD & Applications 2022 [Journal paper](#) 19(6)
- Google Developers [Dev Library](#)

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